

When Fair Enough Is Not Enough: Allocation Policies and Long-Term Benefit Gaps

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May 2026

Abstract—This project reviews three papers that study long-term fairness in algorithmic decision-making. Zezulka and Genin (2024) [1] focus on the policy that translates predictions into decisions. Scher et al. (2023) [2] focus on the dynamics of the surrounding labor market. Xu et al. (2024) [3] focus on how to aggregate group-level statistics across time. Reading them together clarifies that long-term fairness depends on the policy, the environment, and the aggregation rule used to summarize outcomes.

I. MOTIVATION

The use of ML and AI models has become common in decision-making settings where the outcome of the decision impacts the life of the person it affects. Public Employment Services use risk scores to decide who receives access to training programs. Banks use them to decide who receives a loan. Universities use them to decide who is admitted. The shared concern across these settings is that the standard tools of algorithmic fairness evaluate the model at a single point in time, before deployment, against criteria such as statistical parity or equal opportunity. These criteria treat fairness as a property of the predictor in isolation. Once the model is deployed, however, decisions begin to change the world the model observes: people enter or leave the system, skills improve or stagnate, and future training data reflects the choices of past models. A model that satisfies a static fairness criterion at training time can still produce outcomes that widen group gaps over time. The three papers that I reviewed in this survey share this concern from different angles and propose ways to make fairness evaluation account for what happens after deployment.

II. INTRODUCTION

The three papers agree on the diagnosis. Machine learning models trained on historical data inherit past biases, and static fairness metrics evaluated on a single snapshot are insufficient to assess long-term fairness, which depends on multiple factors. They focus their analyses in different factors. Zezulka and Genin focus on the *policy* that translates predictions into decisions, and ask whether fairness constraints on the predictor improve or worsen the distribution of social goods that the policy produces. Scher et al. focus on the *environment*, building a dynamic model of a labor market in which a public agency intervenes repeatedly, and ask how the interaction between the prediction model and the labor market evolves. Xu et al. focus on the *aggregation rule*, asking how step-wise fairness measurements should be combined across time

so that the resulting long-term metric remains consistent with the standard static notions. Reading the three papers together clarifies that long-term fairness is not a single object.

III. ABOUT THE THREE PAPERS

A. Zezulka and Genin (2024): *Fair Predictions vs. Fair Distribution of Social Goods* [1]

1) *Research problem and motivation*: Public Employment Services (PES) have started to introduce ML systems to predict who, among the recently unemployed, is likely to remain in long-term unemployment (LTU), and use those predictions to target labor market programs such as vocational training. The authors observe that the algorithmic fairness literature has concentrated on the distribution of predictions at training time and argue that this is the wrong object of analysis. What is just or unjust is how the predictions are used (the policy) and the outcome of that decision. The same risk score can support completely different policies. For example, *Policy doves*, associated with the Belgian PES, prioritize individuals at the highest predicted risk and target them with training. On the other hand, *Policy hawks*, associated with the Austrian AMS system, treat the high-risk group as hopeless cases and concentrate help on the middle of the distribution (the 30–70th percentiles), based on efficiency. The paper asks which of these policies produce better long-term distribution of social goods and whether fairness constraints on the score change the answer.

2) *Methodology and approach*: The pipeline is run on the Swiss Active Labor Market Policy (ALMP) Evaluation Dataset, with 64,296 individuals. It has four stages. (i) Using double-robust ML, the authors estimate the effect of each of six programs (vocational training, computer, language, job search, employment, personality) on the probability of LTU, with “no program” as the baseline, yielding an Individualized Average Treatment Effect (IATE) per program per person. (ii) They estimate three risk scores by logistic ridge regression: one unconstrained, one constrained by statistical parity, and one by equal opportunity. All three achieve 64–65% accuracy at threshold 0.5. (iii) For each risk score they build two priority lists, the Belgian (dove) list in decreasing order of risk, and the Austrian (hawk) list which prioritizes the 30–70th percentiles. (iv) For each priority list they assign people to programs under two schemes: random assignment among programs with available capacity, and optimal assignment, which sends each person to the program with the highest estimated effectiveness

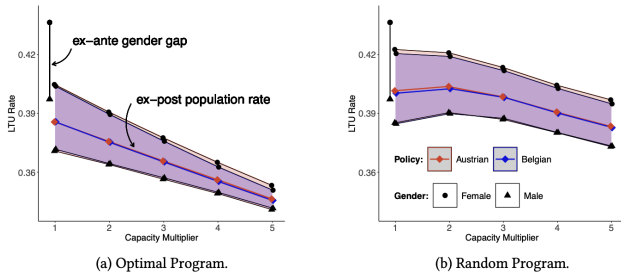


Fig. 1. Long-term unemployment rate and gender gap as a function of program capacity, under the Austrian (hawks) and Belgian (doves) policies. Adapted from [1].

for them based on the IATEs. They also study the effect of multiplying the agency capacities.

3) *Key results*: Three findings stand out. First, requiring the risk score to satisfy statistical parity or equal opportunity *increases* the gender reemployment gap. The mechanism is that the fairness constraint flattens the risk distribution of women toward the men’s distribution, which underestimates women’s true LTU risk and reserves fewer seats in effective programs for them. At baseline program sizes, the Belgian policy with optimal assignment produces a 3.2% gender gap under unconstrained scores and a 4.1% gap under equal opportunity (Fig. 1). When capacity is multiplied by five, the unconstrained policy shrinks the gap, while equal opportunity barely moves it. Second, the dove policy is at least as efficient as the hawk policy: lower overall LTU and smaller gender gap. The supposed efficiency justification for the Austrian approach does not survive a long-term distributive evaluation. Third, individualized assignment based on estimated treatment effects beats random assignment regardless of the priority list, reducing LTU by 1.5 percentage points at baseline capacity and 3.7 points at fivefold capacity.

4) *Strengths and limitations*: As strengths, the analysis is grounded in administrative data, separates the pipeline into clean stages so the source of each effect can be traced, and uses estimated treatment effects rather than synthetic outcomes. The authors are explicit about their own limitations. They do not test all fairness constraints, do not quantify uncertainty in the IATEs, and do not consider online assignment. They describe the approach as paternalistic since it does not incorporate the preferences of the unemployed themselves, and note that the reliance on risk scores may itself be an unnecessary detour relative to directly optimizing for distributive outcomes.

5) *Significance and contribution*: The conceptual contribution is the notion of *prospective fairness*: a fair deployment requires anticipating the causal effect of the algorithm on the distribution of outcomes. The empirical contribution is showing, on real data, that prediction-level fairness constraints can be counterproductive at the level of social goods distribution. This shifts the unit of analysis from the predictor to the policy that uses it. Two concepts used in this paper were used as inspiration for my experiment: allocation policies (hawks vs. doves) and incremental capacity.

B. Scher et al. (2023): Long-Term Fairness Dynamics of Targeted Help [2]

1) *Research problem and motivation*: Scher et al. take the surrounding labor market itself. Their motivation is that a PES does not act on a static dataset: it intervenes repeatedly on a population whose skills, composition, and outcomes evolve over time. Job seekers enter and leave the pool, skills grow when individuals receive help, and the historical data used to train the prediction model is itself a product of past decisions. The authors argue that without an explicit dynamic model of this loop, claims about long-term fairness cannot be tested. The paper is inspired by the Austrian AMS system, which classifies job seekers into three prospect groups (low, moderate, high) and concentrates aid on the moderate group on efficiency grounds. The question Scher et al. ask is what happens to group inequality once such a system is allowed to run over time inside a labor market that has its own internal mechanics.

The conceptual contribution of the paper is to treat the labor market as a dynamic system with three components: a replenishing pool of job seekers, a probabilistic labor market that decides who finds a job at each timestep, and a PES that intervenes through a prediction model. Skills are not fixed: they grow when individuals receive help and the rate of growth depends on the type of help received. Individuals leave the pool either by finding a job or by exceeding a maximum unemployment time T_u^{max} , after which they drop out of the system automatically. New job seekers enter from a background population whose skill distribution differs across groups. This setup makes explicit that any fairness assessment depends on three things at once: the prediction model, the policy that uses it, and the market dynamics that determine who finds a job.

2) *Methodology and approach*: The model is agent-based and operates on a synthetic population. Each individual has two skill features x_1, x_2 and a binary protected attribute x_{pr} . The real skill is

$$s_{real}(x_1, x_2) = \frac{1}{2}(x_1 + x_2), \quad (1)$$

where x_1 is independent of x_{pr} and x_2 is shifted by a parameter α_{pr} so the privileged group has higher mean. The labor market is modeled by a logistic probability of finding a job at each timestep,

$$P(\text{job} | s_{real}) = \frac{1}{1 + e^{-(\alpha_1 s_{real} - \beta_1 + b)}}, \quad (2)$$

where the bias term $b = \beta_b(x_{pr} - 0.5)$ is zero in an unbiased market and positive in a biased one. The PES does not observe x_2 . It compares two prediction models. The base model uses only x_1 . The full model uses x_1 together with the protected attribute x_{pr} as a predictor, which boosts accuracy on average but uses group membership as an input. Individuals classified as low-prospect are placed in a waiting group for ΔT_u timesteps, where their skills are updated by

$$x_i^{t+1} = \max[x_i^t + k(x_i^{max} - x_i^t), x_i^t], \quad i = 1, 2, \quad (3)$$

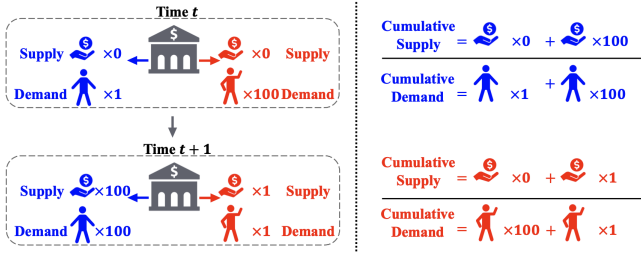


Fig. 3. Loan example illustrating ratio-before vs. ratio-after aggregation. Step-wise aggregation reports zero bias across the two timesteps, while the Long-term Benefit Rate correctly flags the trajectory as biased. Adapted from [3].

$$\eta_g^D(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t D_g(s_t, a_t) \right]. \quad (8)$$

The *Long-term Benefit Rate* of group g is the ratio $\eta_g^S(\pi)/\eta_g^D(\pi)$, and the *bias* of a policy is the maximum gap of this rate across groups,

$$b(\pi) = \max_{g \in G} \frac{\eta_g^S(\pi)}{\eta_g^D(\pi)} - \min_{g \in G} \frac{\eta_g^S(\pi)}{\eta_g^D(\pi)}. \quad (9)$$

This is a *ratio-after-aggregation* metric: supply and demand are summed over time first and only then divided. Going back to the loan example, the bias becomes $|100/101 - 1/101| = 99/101$, which correctly flags the trajectory as unfair. The framework recovers the standard static fairness notions as the special case of time horizon one: by choosing S_g and D_g appropriately, ELBERT (Equal Long-term Benefit Rate) covers demographic parity, equal opportunity, equalized odds, accuracy parity, and equality of discovery probability. The constrained learning problem is

$$\max_{\pi} \eta(\pi) \quad \text{s.t.} \quad b(\pi) \leq \delta. \quad (10)$$

They highlight that the technical obstacle is that the bias term is a difference of *ratios* of cumulative rewards, which does not satisfy a Bellman equation directly, so standard policy optimization methods do not apply. The authors prove that the policy gradient of the Long-term Benefit Rate reduces analytically to the standard reinforcement learning policy gradient through a *fairness-aware advantage function*. This makes standard policy optimization methods such as PPO (Proximal Policy Optimization) usable for bias mitigation. They propose an algorithm called ELBERT Policy Optimization (ELBERT-PO) which optimizes a regularized objective of the form $J(\pi) = \eta(\pi) - \alpha b(\pi)^2$.

3) *Key results*: ELBERT-PO is evaluated on three benchmarks: credit approval for lending, infectious disease control over a population network, and attention allocation for incident monitoring. Across the three, it produces the lowest bias among three baseline policy optimization algorithms while keeping utility high. The mechanism for bias reduction differs by environment: in lending the disadvantaged group's supply increases while its demand decreases, in infectious disease control the supplies are rebalanced with demand mostly fixed, and in attention allocation demand itself shifts. The authors

note explicitly that ELBERT, like the static notions it generalizes, controls only the supply-demand ratio and not the absolute level of demand. In cases where the absolute level matters (e.g., the number of qualified applicants in lending), an extra regularization term on $\eta_g^D(\pi)$ can be added.

4) *Strengths and limitations*: The strength is generality: a single MDP-based definition recovers most standard fairness notions and fixes the temporal discrimination pathology of step-wise aggregation. The limitation acknowledged by the authors is the focus on the ratio rather than the absolute level of demand, which can be problematic in applications where the demand is itself a substantive quantity of interest.

5) *Significance and contribution*: The contribution is twofold. Conceptually, the supply-demand MDP and the Long-term Benefit Rate provide a unified language for long-term group fairness, with the ratio-after-aggregation property as a principled design choice. Algorithmically, the reduction of the policy gradient to standard form makes the framework practically usable with mainstream RL methods. From this paper the idea of long-term aggregation metrics was used as inspiration for my experiment.

IV. SYNTHESIS

The three perspectives are not in conflict. A serious long-term fairness evaluation in a real deployment needs to specify all three: the policy that maps predictions to decisions, the environment that maps decisions to future states, and the rule that aggregates outcomes into a fairness summary. Table I summarizes the comparison.

V. EXPERIMENT

My experiment synthesizes ideas from the papers into a small employment support simulation. Zezulka and Genin motivate the policy question: the same risk score can generate different distributions of social goods depending on how it is used. Scher et al. motivate the dynamic question: repeated interventions change the population over time. Xu et al. motivate the measurement question: long-term fairness should compare cumulative benefit with cumulative demand, rather than only averaging step-wise gaps.

I use the ACS Employment task as a proxy for employment-support allocation, not as a direct model of a public employment service. Employment status is used to define observed need: people who are not employed are treated as needing support. A logistic regression model estimates employment probability, and unemployment risk is defined as its complement. The sensitive attribute in the main experiment is sex, coded as Female and Male, and it is excluded from the prediction model. The technical details of the data, model, policies, simulation updates, and metrics are reported in Appendix A.

The experiment has two parts. The static part evaluates one allocation round under a fixed support budget. The dynamic part repeats the allocation process over time, allowing support and non-support to change future employment probabilities. This structure makes it possible to compare policies that look similar in a single round but produce different cumulative benefit patterns.

TABLE I
SUMMARY OF THE THREE PAPERS.

Paper	Focus	Setting	Key concepts	Main finding
Zezulka & Genin (2024)	Policy	Swiss PES, real administrative data, six labor market programs	Prospective fairness, doves vs. hawks, Optimal program assignment	Fairness constraints on the risk score widen the gender reemployment gap, the dove policy is at least as efficient as the hawk, increases in capacity lead to smaller gaps, and optimal program assignment decreases LTU faster.
Scher et al. (2023)	Environment	Synthetic agent-based labor market with PES intervention	Labor market as a dynamic system: evolving skills, in-flow and outflow of job-seekers, feedback between PES decisions and future data	Trade-off between BGSD and counterfactual fraction; targeting has limited impact unless aid is concentrated on the high-prospect group.
Xu et al. (2024)	Aggregation over time rule	Lending, infectious disease control, attention allocation	Supply-Demand, MDP, Long-term Benefit Rate (ratio after aggregation), policy optimization	ELBERT-PO reduces long-term bias while maintaining utility and avoids the temporal-discrimination pathology of step-wise aggregation.

A. Static Allocation

The static experiment compares three allocation rules. Utility only allocation rule selects individuals with the highest unemployment risk. Demographic parity allocates support so that Female and Male groups receive approximately the same overall selection rate. Equal opportunity is included as an idealized benchmark: it uses observed need (employment label) directly and equalizes selection among individuals who are not employed. This rule is not meant to be fully deployable in the same way as the others, because true need may not be observed cleanly in real settings. It is used as a reference point for asking whether support reaches those who need it.

Each rule is evaluated under two targeting strategies. The high-risk strategy prioritizes the largest unemployment-risk scores. The middle-risk strategy focuses on individuals in the middle of the risk distribution, inspired by systems that concentrate resources on people considered more likely to benefit from intervention. The comparison separates two questions: whether support is distributed evenly across groups, and whether support reaches individuals with observed need.

B. Dynamic Simulation

The dynamic simulation repeats the allocation logic across rounds. In each round, currently unemployed individuals are eligible for support. A policy selects a subset of eligible individuals under the available budget. Supported individuals receive a positive update to their employment probability, and eligible individuals who are not supported experience a small decay. The next employment status is then simulated from the updated probability.

Fig. 4 summarizes this logic. Individuals who find employment exit the active pool and are replaced. Individuals who remain not employed continue to the next round and accumulate unemployment duration. If this duration reaches $T_{\max}$, they exit through timeout.

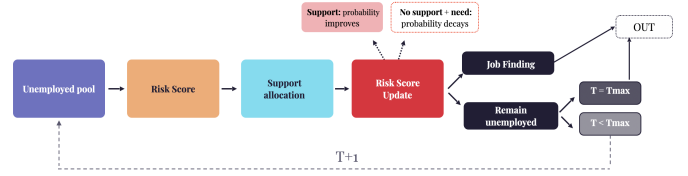


Fig. 4. Conceptual diagram of the dynamic simulation. Support decisions update employment probabilities, which determine whether individuals find employment, remain unemployed, or exit after reaching the maximum unemployment duration.

The dynamic experiment compares utility only and demographic parity with a long-term-aware heuristic. The long-term-aware rule gives additional priority to the group with the lower cumulative benefit rate in previous rounds. It is not an optimized reinforcement-learning policy. Its purpose is simpler: to test whether explicitly feeding cumulative benefit gaps back into the allocation rule changes long-term disparities.

C. Results and Interpretation

The baseline pattern is important for interpreting the results. In the test population, men have a higher average predicted unemployment risk, while women have a higher observed need rate. Therefore, predicted risk and observed need do not rank the two sex groups in exactly the same way. This makes the allocation problem nontrivial: a policy that follows the risk score can differ from a policy that follows observed need.

1) *Static allocation*: Figure 5 shows the allocation results under high-risk and middle-risk targeting. Under high-risk targeting, utility only selects a larger share of men because it follows the prediction pattern. Demographic parity reduces the selection gap by forcing similar selection rates across sex groups. Equal opportunity behaves differently: instead of equalizing selection over the whole population, it equalizes selection among individuals with observed need.

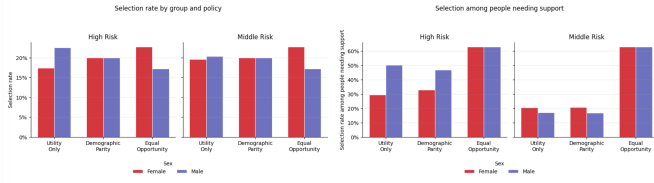


Fig. 5. Static allocation results by policy, sex group, and targeting strategy. The left panel reports overall selection rates, while the right panel reports selection among individuals needing support.

The main interpretation is that the policies solve different fairness problems. Utility only allocation asks who has the highest predicted risk. Demographic parity asks whether both groups receive support at similar overall rates. Equal opportunity asks whether support reaches people with observed need at similar rates. These goals overlap, but they are not equivalent. In this experiment, they differ because the group with higher predicted risk is not the same as the group with higher observed need.

Middle-risk targeting changes the selected population. It lowers the average risk profile of supported individuals and makes the allocation less focused on observed need. This can look more moderate from an efficiency perspective, but it can also move support away from the individuals with highest need. This is the static version of the hawk vs dove problem discussed by Zezulka and Genin: the definition of the target population is itself a relevant choice in terms of fairness measurement.

2) *Budget as a policy choice*: Figure 6 shows that budget capacity changes who receives support. As capacity increases, more people can be selected and the mean risk among selected individuals decreases. Under high-risk targeting, this is expected: once the highest-risk cases have been selected, additional capacity reaches lower-risk individuals. Under middle-risk targeting, support remains centered on the middle of the distribution even when more capacity is available.

One non-obvious result is that, under high-risk targeting, utility only allocation produces the smallest group gap in mean selected risk. This happens because the risk score itself already ranks men higher on average, while observed need is higher among women. Under middle-risk targeting, equal opportunity can increase the gap at higher capacities because the rule follows observed need rather than the risk distribution.

This result matters because budget is not only a technical parameter. Larger budgets reduce scarcity, but they do not remove the consequences of the priority rule. Capacity determines how many people can be reached, targeting determines who is reached first.

3) *Dynamic simulation*: The dynamic results show that one-round allocation fairness does not fully determine cumulative benefit. Table II reports final-round averages across simulation seeds. In this implementation, demographic parity produces the smallest benefit-rate gaps under both targeting strategies, with the smallest gap under middle-risk targeting. Utility only high-risk targeting produces the largest benefit

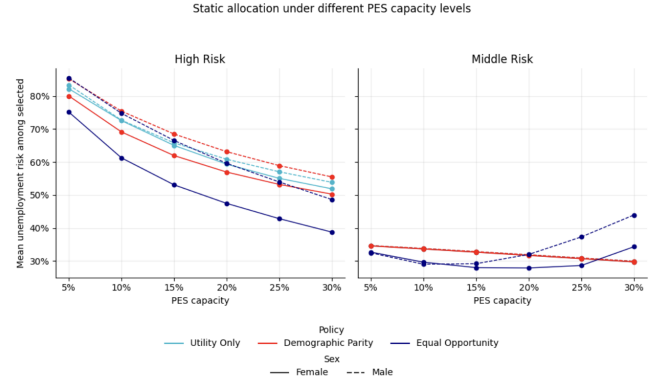


Fig. 6. Static allocation under different budget levels. Larger budgets expand access, but the targeting rule determines which part of the risk distribution is reached first.

TABLE II
FINAL-ROUND DYNAMIC RESULTS AVERAGED ACROSS SIMULATION SEEDS. LTA DENOTES THE LONG-TERM-AWARE HEURISTIC AND UO DENOTES UTILITY ONLY POLICY.

Policy variant	Benefit gap	Selection gap	Job finding	Timeout
DP, middle-risk	0.004	0.006	0.522	0.019
DP, high-risk	0.013	0.006	0.548	0.010
LTA, middle-risk	0.008	0.006	0.521	0.019
LTA, high-risk	0.016	0.032	0.549	0.009
UO, middle-risk	0.010	0.010	0.522	0.019
UO, high-risk	0.019	0.035	0.549	0.009

gap.

The comparison also shows a trade-off between targeting strategy and outcome interpretation. High-risk targeting produces higher job-finding rates among individuals needing support, but it also produces larger benefit and selection gaps. Middle-risk targeting produces smaller benefit gaps in some cases, but it has higher timeout rates. This suggests that middle-risk targeting reaches easier cases while leaving more high-need individuals at risk of remaining unsupported long enough to exit through timeout.

The long-term-aware heuristic does not dominate demographic parity in this simulation. The heuristic uses accumulated benefit deficits, but it still operates inside the same risk-based allocation structure. Under high-risk targeting, the adjustment is not enough to offset the baseline difference between risk and need. Under middle-risk targeting, it improves on utility only policy in terms of benefit gap, but demographic parity still produces the smallest cumulative gap.

The support-rate gap follows the same broad pattern as the selection gap, which suggests that differences in who receives support are carried into the cumulative benefit calculation rather than appearing only as a one-round allocation artifact. Another interesting finding is that across variants, the allocation rule itself remains one of the main drivers of the dynamic outcomes.

Overall, the experiment supports the central claim of the paper. Static diagnostics such as the selection-rate gap are useful, but they are incomplete. The same allocation rule can look

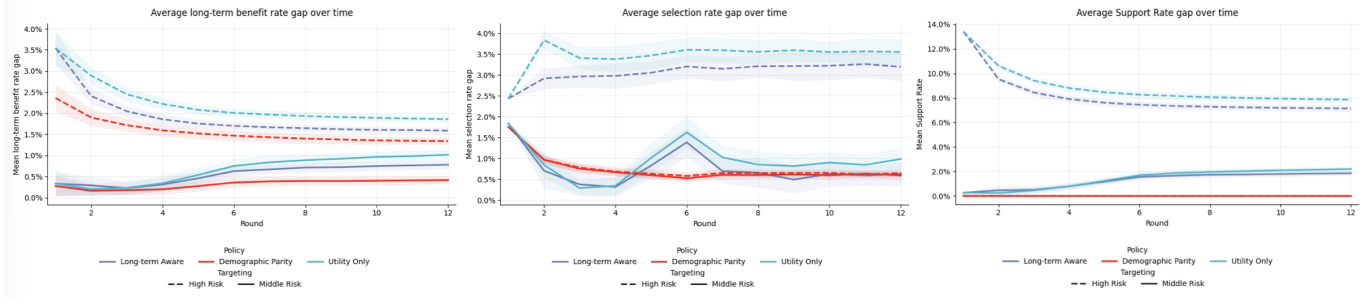


Fig. 7. Dynamic fairness and outcome metrics over time. The figure compares benefit-rate gaps, selection-rate gaps, and support-rate gaps across policies and targeting strategies.

acceptable in one round and still generate different cumulative benefit, job-finding, and timeout patterns over time. Long-term fairness depends on the policy rule, the simulated environment, the target population, the budget, and the aggregation rule used to summarize benefit and demand.

VI. CONCLUSION

This project reviewed three complementary approaches to long-term fairness in algorithmic decision-making. Zezulka and Genin show that fairness cannot be evaluated only at the level of the predictor, because the same score can produce different distributions of social goods depending on the policy that uses it. Scher et al. show that the surrounding environment matters because interventions change the population over time. Xu et al. show that fairness also depends on how group outcomes are aggregated across time, and that cumulative benefit should be compared against cumulative demand rather than evaluated only through step-wise gaps.

In my experiment I applied these ideas to a simplified employment-support setting. The static results show that utility only, demographic parity, and equal opportunity answer different questions. Utility only allocation only follows predicted risk. Demographic parity equalizes overall selection. Equal opportunity checks whether support reaches people with observed need. Because predicted risk and observed need differed by sex, these rules produced different allocation patterns.

The dynamic simulation shows why the static comparison is not enough. Once support affects future employment probabilities, similar selection behavior can lead to different cumulative benefit rates, job finding rates, and timeout patterns. In this simulated setting, demographic parity produced the smallest benefit-rate gaps, while high-risk targeting produced higher job-finding rates and middle-risk targeting produced higher timeout rates. This does not prove that demographic parity is generally the best policy. It shows that the fairness conclusion depends on the data pattern, the target population, the budget, the support effect, and the way benefit is accumulated over time.

The broader lesson is that long-term fairness should be evaluated where benefits and harms accumulate: across policies, environments, and time. Static fairness metrics remain useful

diagnostics, but they do not fully describe the downstream effects of an allocation system.

Future work could extend this framework to longitudinal administrative data, richer employment histories, estimated treatment effects, and additional sensitive attributes. The same ideas could also be adapted beyond employment support, including lending, admissions, health interventions, public benefits, and other settings where algorithmic decisions allocate scarce opportunities over time.

APPENDIX A METHODS

A. Data

This experiment uses the Folktabs ACS Employment task, constructed from U.S. Census ACS Public Use Microdata [4]. The target variable is employment status, where $Y_i = 1$ indicates that individual i is employed and $Y_i = 0$ indicates that the individual is not employed. I define need for support as

$$N_i = 1 - Y_i,$$

so individuals observed as not employed are treated as needing employment support.

The sensitive attribute in the main experiment is sex:

$$A_i \in \{\text{Female}, \text{Male}\}.$$

The prediction model excludes sex and uses the remaining ACS features to estimate employment probability. This keeps the experiment focused on how allocation rules transform a common risk score into different group outcomes.

B. Model

The baseline prediction model is a logistic regression model with preprocessing for numerical and categorical variables. The model estimates the probability of employment:

$$\hat{p}_i = P(Y_i = 1 \mid X_i). \quad (11)$$

The corresponding unemployment risk score is defined as

$$\hat{r}_i = 1 - \hat{p}_i. \quad (12)$$

Individuals with larger values of $\hat{r}_i$ are interpreted as having higher unemployment risk. The goal of my experiment is not to

maximize predictive performance, but to study how different allocation rules transform the same risk score into different short-term and long-term fairness outcomes.

TABLE III
RISK MODEL PERFORMANCE BY SEX GROUP.

Group	n	Accuracy	AUC	Mean $\hat{p}$	True emp.	TPR	FPR
Male	23570	0.786	0.756	0.885	0.725	0.933	0.708
Female	23372	0.677	0.673	0.832	0.638	0.932	0.691

The risk model is reasonably balanced across sex groups in terms of AUC and average predicted employment probability. However, when probabilities are converted into binary predictions using a 0.5 threshold, the model predicts employment for a much larger share of individuals than the observed employment rate. This is reflected in the high false positive rates for both groups. For this reason, the predicted probability is used as a continuous risk score rather than relying on the binary employment classification.

C. Static Fairness

The static experiment evaluates how different allocation policies distribute a fixed support budget at one point in time. For a budget B , each policy selects approximately B percent of the relevant population for support.

The first policy is *utility only* allocation. It selects individuals with the highest unemployment risk:

$$S_i = 1 \quad \text{if } \hat{r}_i \text{ is among the top } B \text{ share of risks.} \quad (13)$$

The second policy is *demographic parity* allocation. It selects the same budget share within each sensitive group:

$$P(S = 1 \mid A = a) \approx B \quad \forall a. \quad (14)$$

The third policy is equal *opportunity allocation*. In this experiment, equal opportunity is implemented as an idealized benchmark because it uses the observed employment label to identify individuals who need support. Among individuals with $N_i = 1$, it allocates support proportionally across groups:

$$P(S = 1 \mid N = 1, A = a) \approx B \quad \forall a. \quad (15)$$

This is only used as a reference for evaluating whether support reaches individuals who actually need it. In a real setting, true need may not be observed in the same way.

For each policy, I evaluate two targeting strategies. The high-risk strategy selects the individuals with the highest unemployment risk. The middle-risk strategy selects individuals whose risk falls within the 30th to 70th percentile range and then prioritizes higher-risk individuals within that middle-risk band. This second strategy is inspired by public employment systems that concentrate resources on individuals considered most likely to benefit from support.

The static metrics are defined as follows. The group selection rate is

$$\text{Selection Rate}_a = \frac{\sum_i \mathbf{1}(S_i = 1, A_i = a)}{\sum_i \mathbf{1}(A_i = a)}. \quad (16)$$

The selection gap is

$$\text{Selection Gap} = \max_a \text{Selection Rate}_a - \min_a \text{Selection Rate}_a. \quad (17)$$

The support rate among individuals who need support is

$$\text{Support Among Need}_a = \frac{\sum_i \mathbf{1}(S_i = 1, N_i = 1, A_i = a)}{\sum_i \mathbf{1}(N_i = 1, A_i = a)}. \quad (18)$$

Finally, the mean unemployment risk among selected individuals is

$$\text{Mean Risk Among Selected}_a = \frac{\sum_i \hat{r}_i \mathbf{1}(S_i = 1, A_i = a)}{\sum_i \mathbf{1}(S_i = 1, A_i = a)}. \quad (19)$$

These metrics show whether each policy selects similar proportions across groups, whether support reaches individuals who need it, and whether selected individuals have comparable risk profiles across groups.

D. Dynamic Simulation

The dynamic simulation starts from the same test population used in the static experiment. In each round, only individuals who are currently not employed are eligible for support. Therefore, the dynamic budget is interpreted as a share of the currently high-need population, not as a share of the full test population.

For each eligible individual, the current unemployment risk is

$$r_{i,t} = 1 - p_{i,t},$$

where $p_{i,t}$ is the current simulated probability of employment. A policy selects a subset of eligible individuals for support. Supported individuals receive a positive update to their employment probability:

$$p'_{i,t} = p_{i,t} + e_{i,t},$$

where $e_{i,t}$ is drawn from the simulated support-effect distribution. Individuals who need support but do not receive it experience a small decay:

$$p'_{i,t} = p_{i,t} - d.$$

The updated probability is clipped to the interval $[0, 1]$, and the next employment outcome is drawn as

$$Y_{i,t+1} \sim \text{Bernoulli}(p'_{i,t}).$$

The metrics below separate allocation fairness, realized benefit fairness, and outcome utility.

1) *Dynamic Metrics*: The main dynamic metric is the long-term benefit rate. For each group a , cumulative demand is the number of times individuals in that group needed support:

$$D_{a,T} = \sum_{t=1}^T \sum_i \mathbf{1}(A_i = a, N_{i,t} = 1). \quad (20)$$

Cumulative benefit is the number of times individuals in that group needed support, received support, and became employed:

$$B_{a,T} = \sum_{t=1}^T \sum_i \mathbf{1}(A_i = a, N_{i,t} = 1, S_{i,t} = 1, Y_{i,t+1} = 1). \quad (21)$$

The long-term benefit rate is

$$BR_{a,T} = \frac{B_{a,T}}{D_{a,T}}. \quad (22)$$

The long-term benefit rate gap is

$$\text{Benefit Rate Gap}_T = \max_a BR_{a,T} - \min_a BR_{a,T}. \quad (23)$$

This metric follows a ratio-after-aggregation logic: benefits and demand are accumulated first, and the group rate is computed afterward. This avoids treating each round as a separate static snapshot and is consistent with long-term fairness metrics that compare cumulative group benefit over cumulative group demand [3].

I also report the cumulative support rate:

$$SR_{a,T} = \frac{\sum_{t=1}^T \sum_i \mathbf{1}(A_i = a, S_{i,t} = 1)}{D_{a,T}}. \quad (24)$$

The support rate gap is

$$\text{Support Rate Gap}_T = \max_a SR_{a,T} - \min_a SR_{a,T}. \quad (25)$$

The job-finding rate among people needing support is

$$JFR_{a,t} = \frac{\sum_i \mathbf{1}(A_i = a, N_{i,t} = 1, Y_{i,t+1} = 1)}{\sum_i \mathbf{1}(A_i = a, N_{i,t} = 1)}. \quad (26)$$

The job-finding gap is

$$\text{Job-Finding Gap}_t = \max_a JFR_{a,t} - \min_a JFR_{a,t}. \quad (27)$$

I also report the total employment rate in each round:

$$\text{Employment Rate}_t = \frac{1}{n} \sum_i Y_{i,t}. \quad (28)$$

Finally, I report the timeout rate after the timeout threshold is reachable:

$$\text{Timeout Rate}_t = \frac{1}{n} \sum_i \text{Timeout}_{i,t}, \quad t \geq T_{\max}. \quad (29)$$

Together, these metrics separate three outcomes: allocation fairness, measured by support and selection gaps; realized benefit fairness, measured by the long-term benefit rate gap; and system-level utility, measured by employment and job-finding outcomes.

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